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(54) **LOW POWER VOLTAGE CONTROLLABLE OSCILLATOR WITH RAIL-TO-RAIL OUTPUT**

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(Continued)

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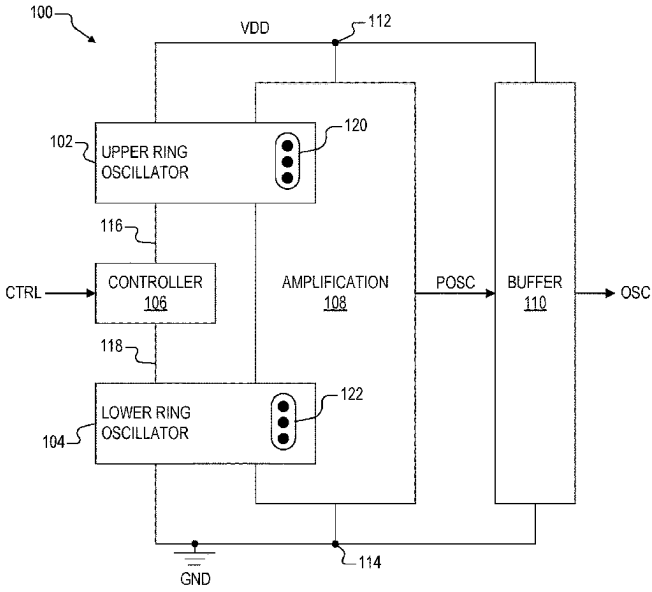
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(57) **ABSTRACT**

A controllable oscillator including an upper oscillator coupled between an upper supply voltage and an upper intermediate node that provides at least one upper oscillating signal on at least one upper oscillating node, a lower oscillator coupled between a lower intermediate node and a lower supply voltage that provides at least one lower oscillating signal on at least one lower oscillating node, an oscillation controller coupled between the upper and lower intermediate nodes, and amplification circuitry coupled between the upper and lower supply voltages, having at least one upper input coupled to the at least one upper oscillating node, having at least one lower input coupled to the at least one lower oscillating node, and having a primary output node for providing a primary rail-to-rail oscillation signal. A coupling circuit may be coupled between one or more upper and lower oscillating nodes for synchronization.

20 Claims, 9 Drawing Sheets



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USPC 331/57

See application file for complete search history.

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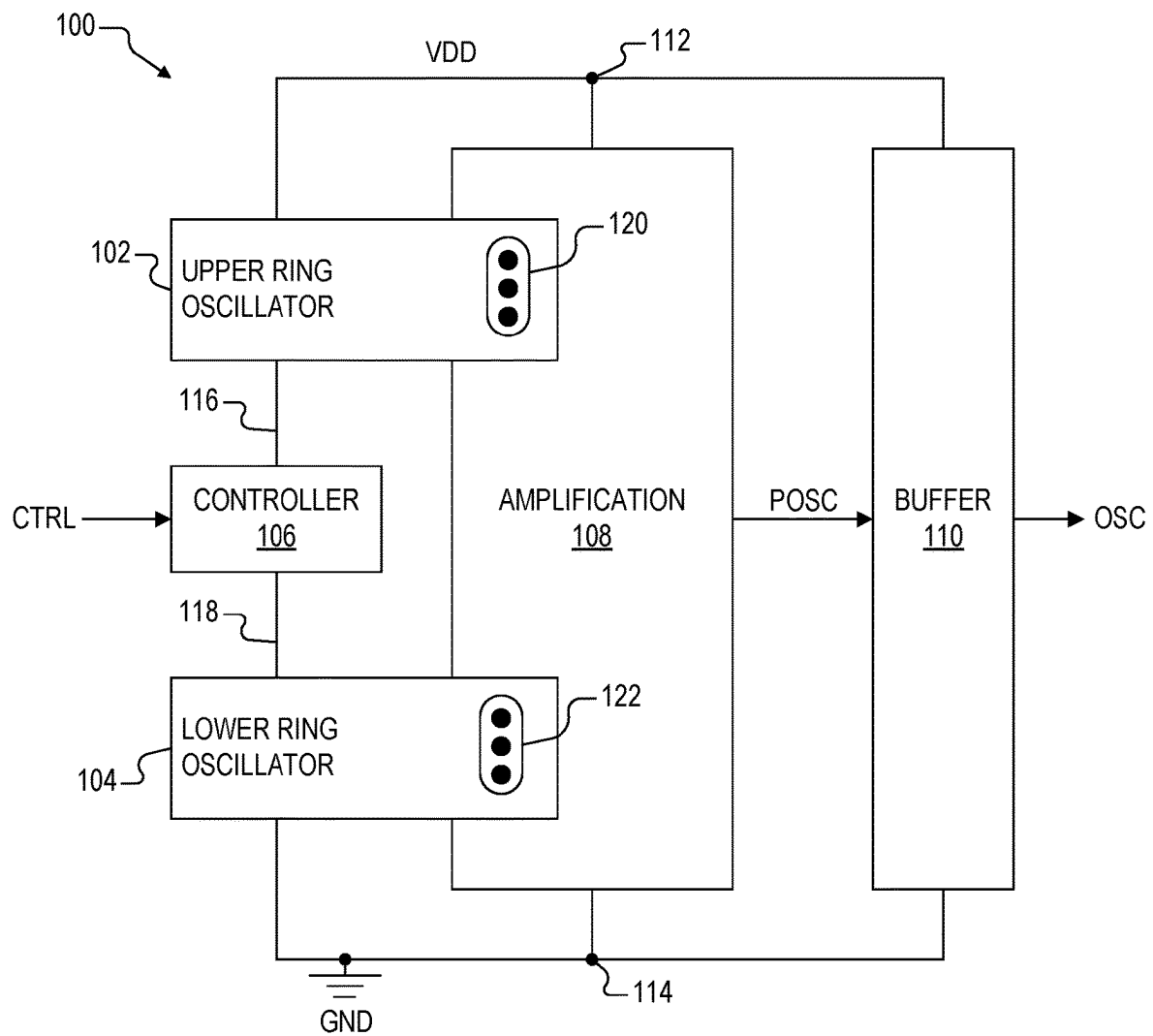


FIG. 1

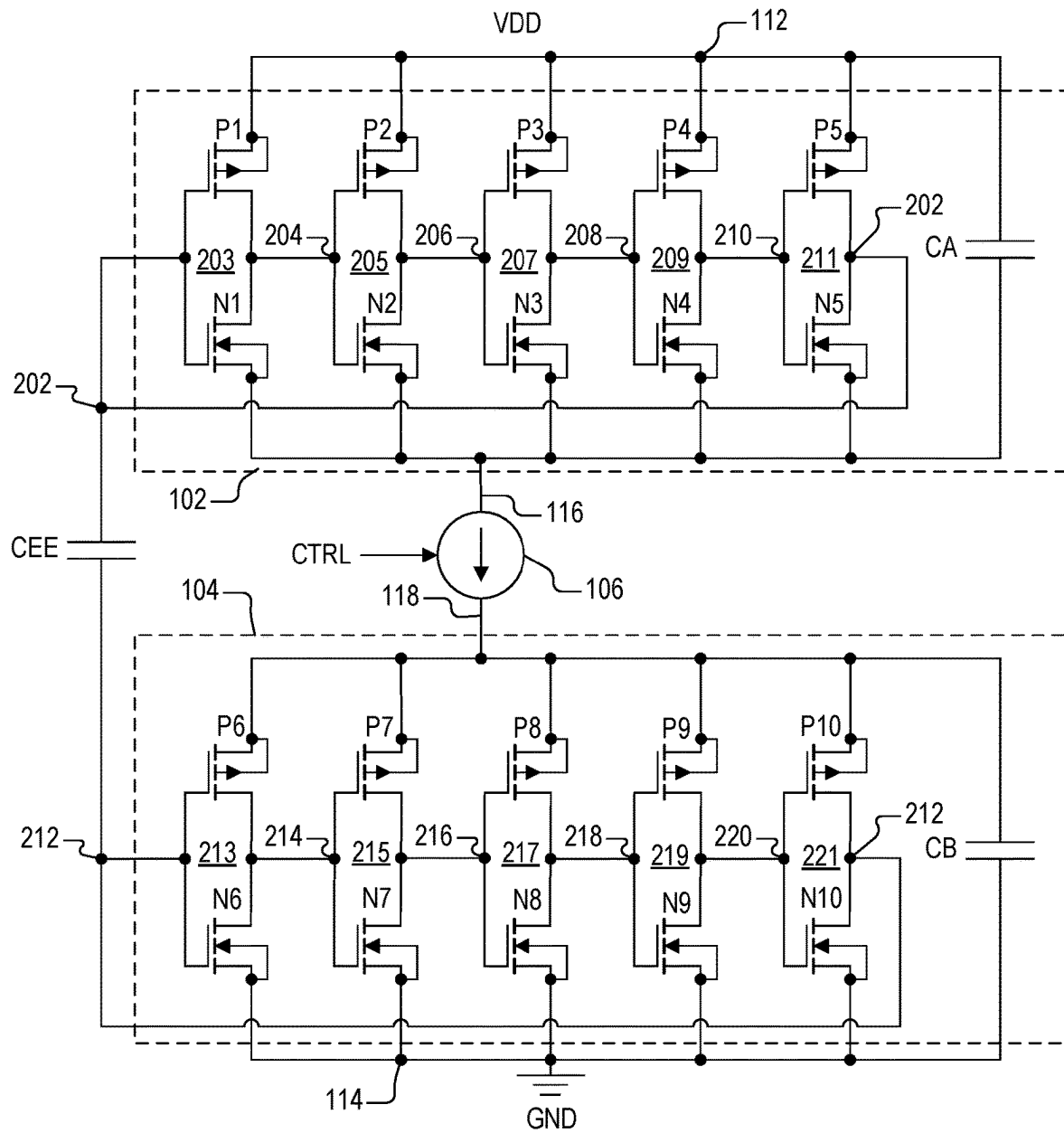


FIG. 2

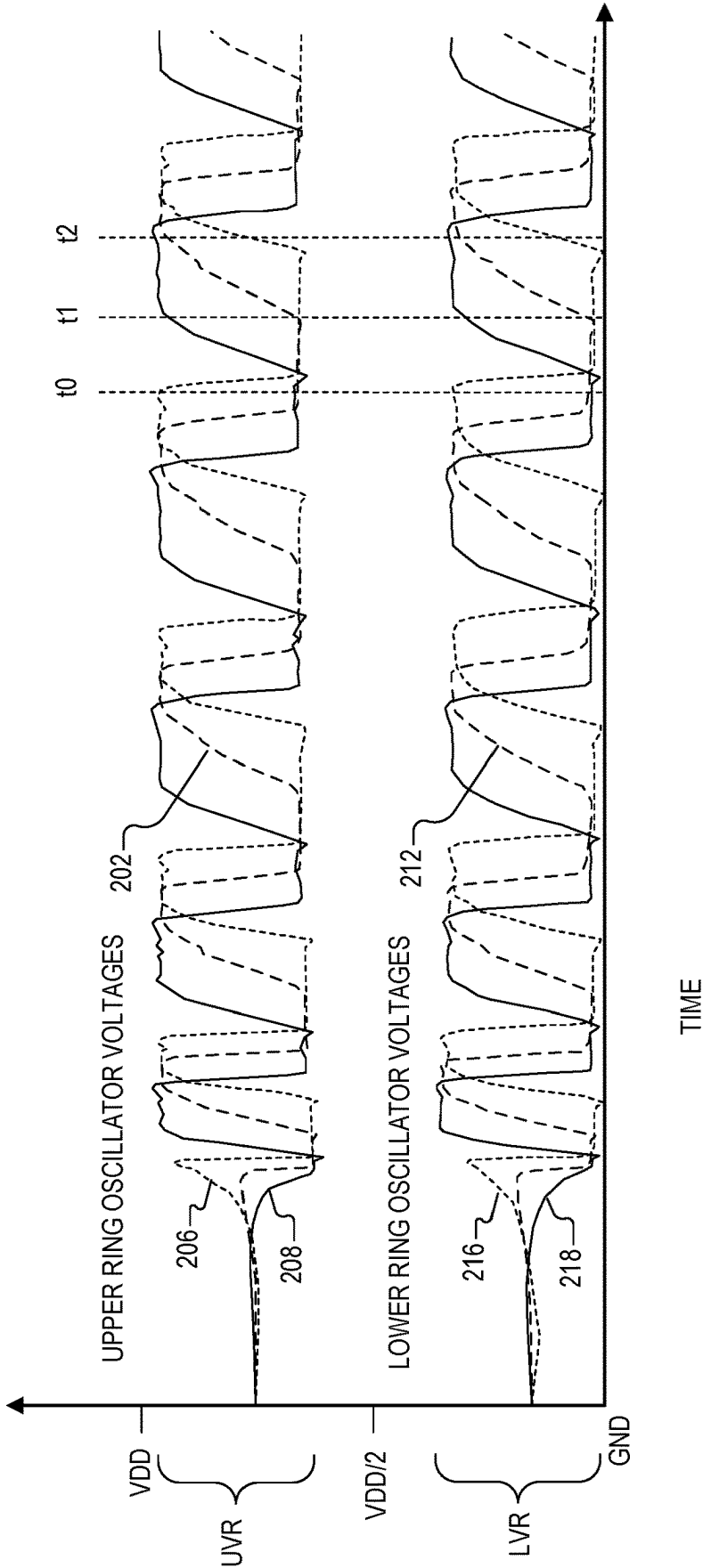


FIG. 3

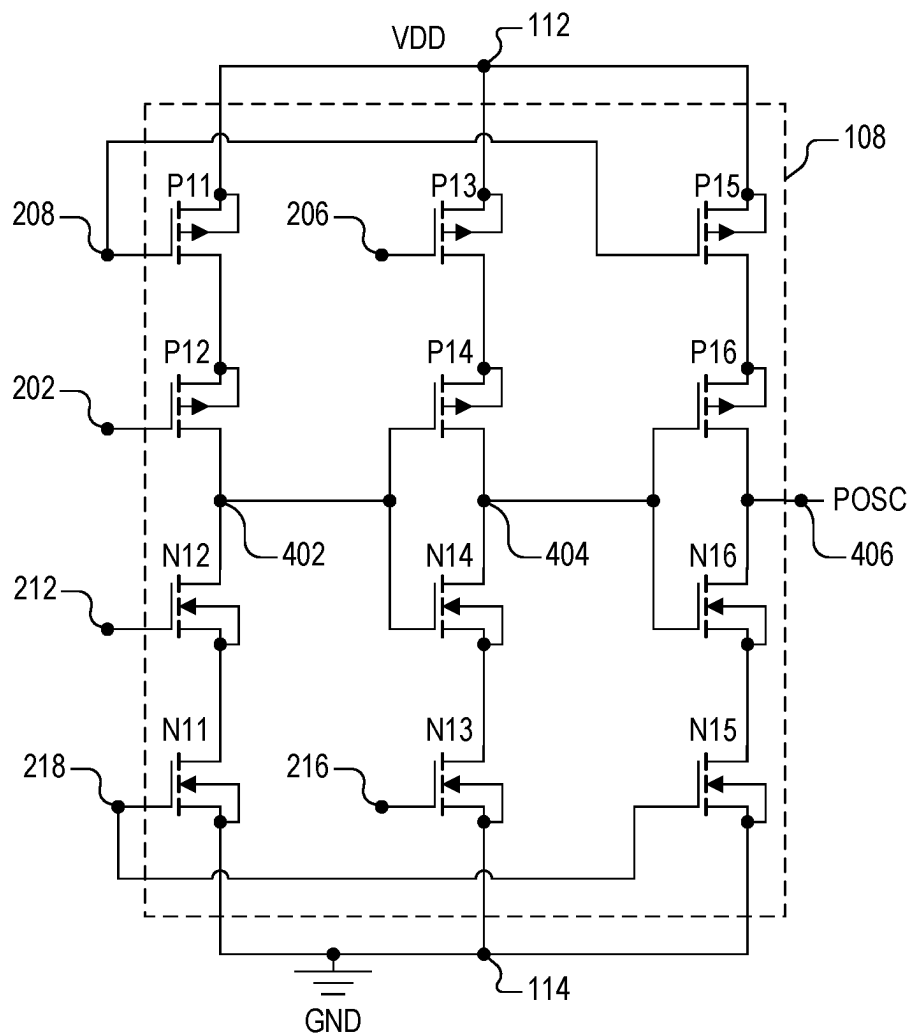
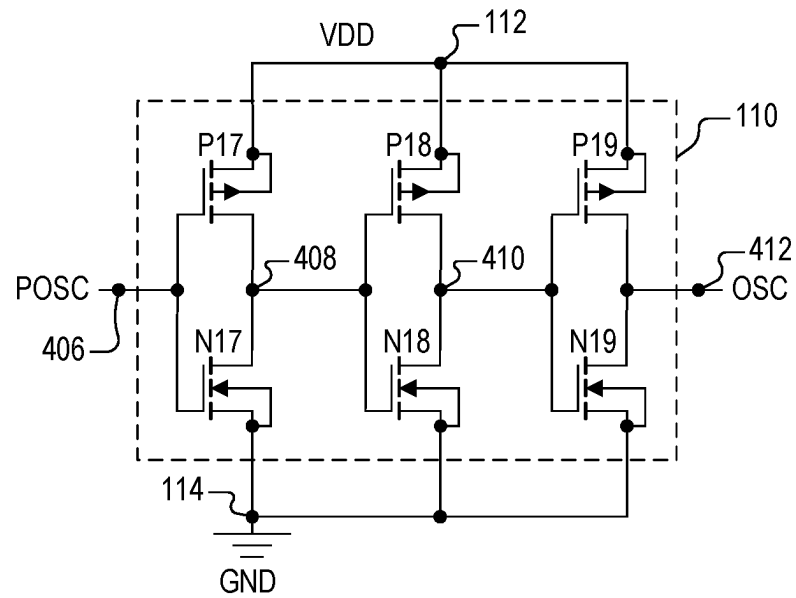


FIG. 4

*FIG. 5*

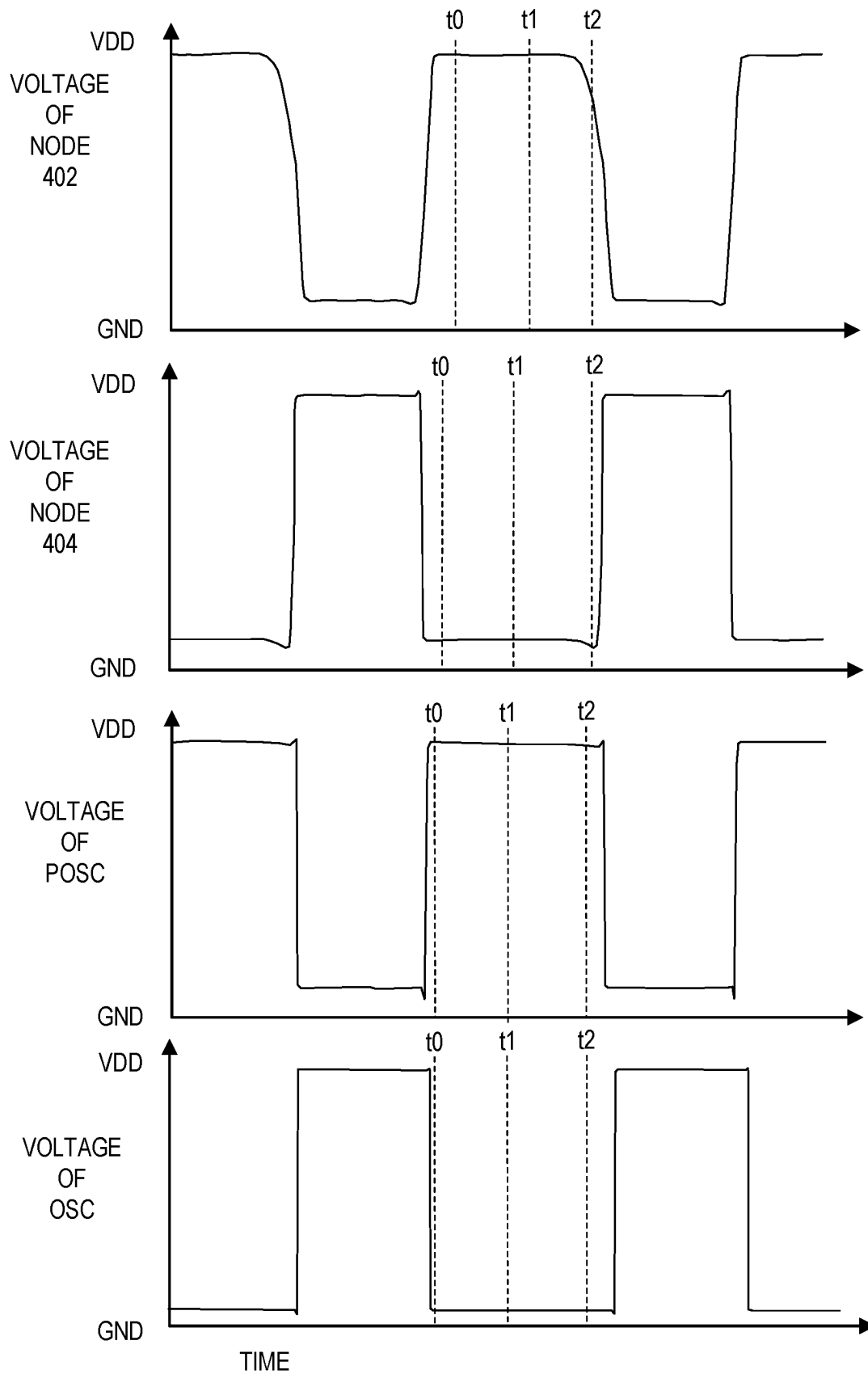


FIG. 6

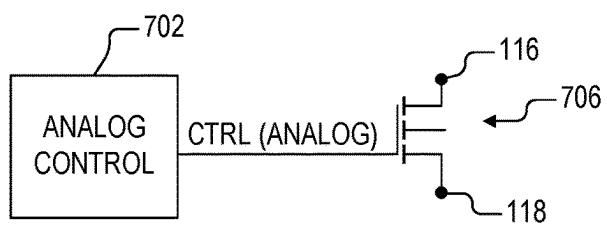


FIG. 7

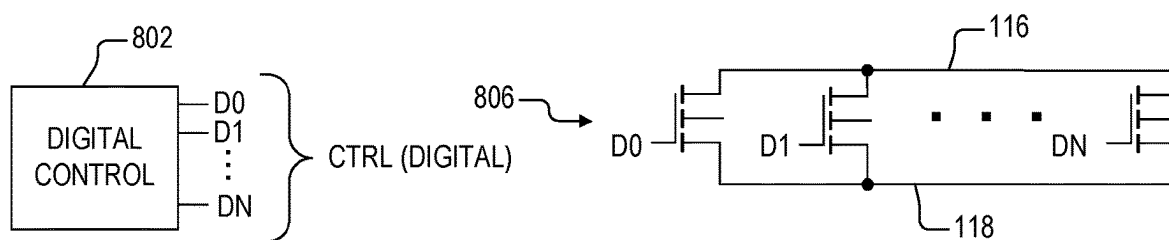


FIG. 8

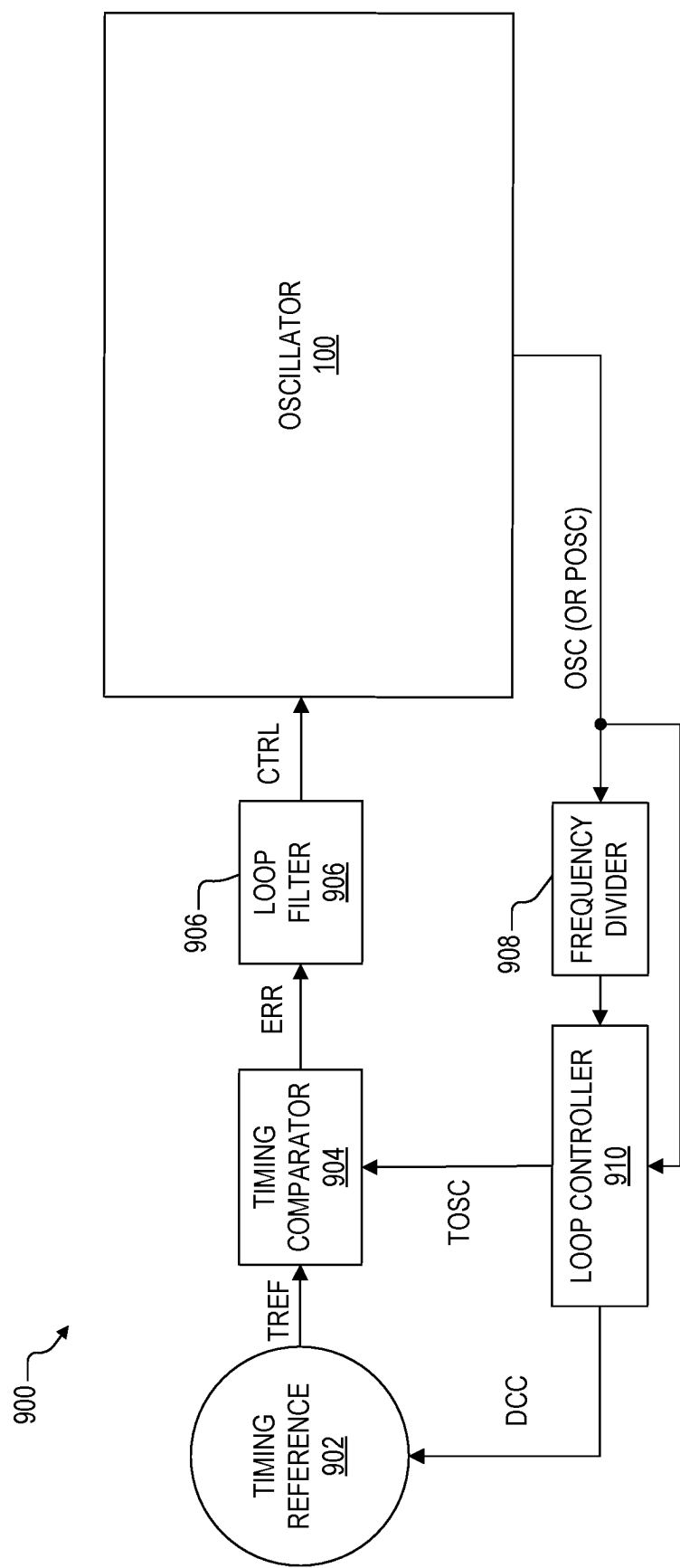


FIG. 9

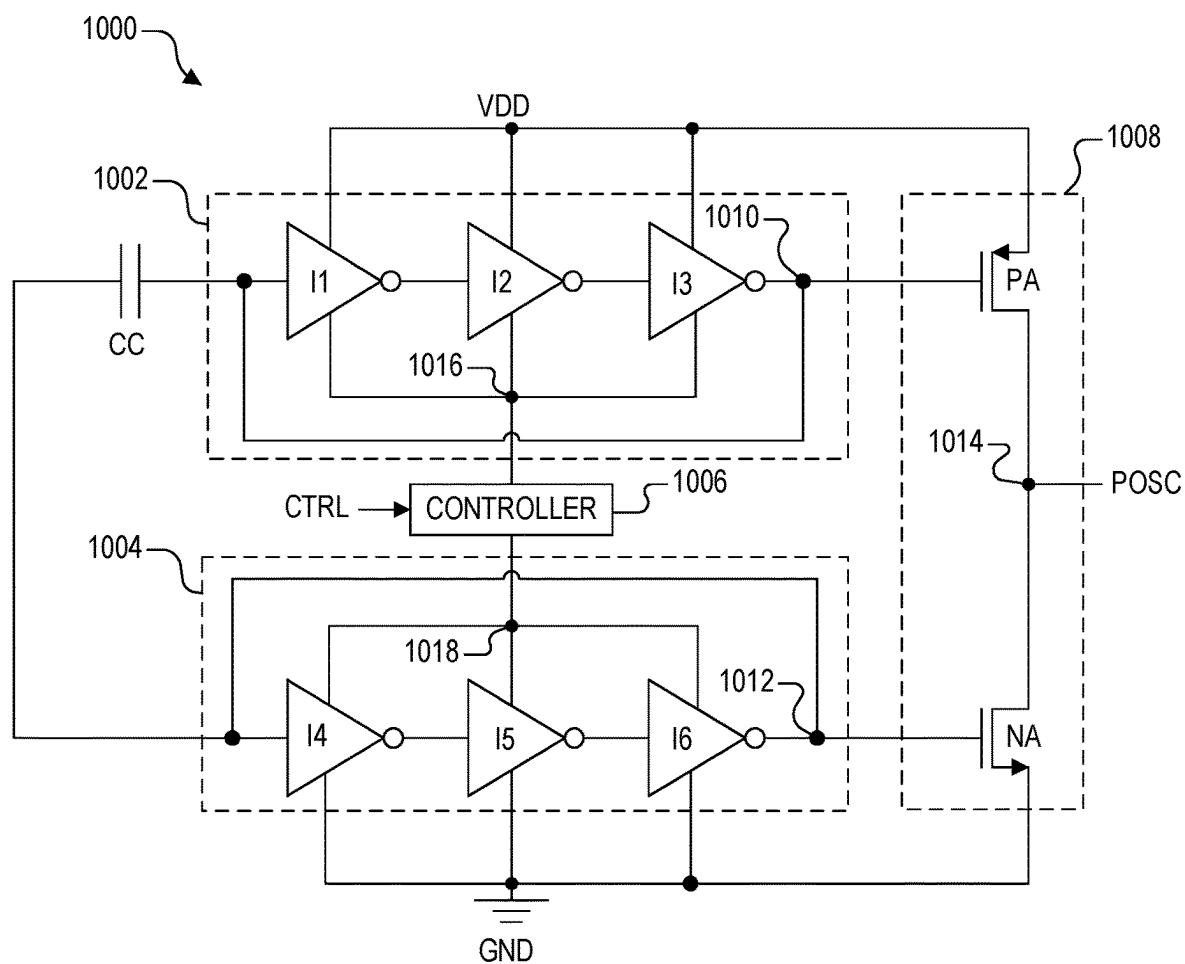


FIG. 10

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LOW POWER VOLTAGE CONTROLLABLE OSCILLATOR WITH RAIL-TO-RAIL OUTPUT

BACKGROUND

Field of the Invention

The present invention relates in general to oscillators, and more particularly to a low power controllable oscillator that provides a rail-to-rail output signal.

Description of the Related Art

Clock generation is a widespread need in any mixed-signal or digital system. Clock signals can be used, for example, to clock a digital processing circuit, to sample analog signals in an analog-to-digital converter (ADC) or a digital-to-analog converter (DAC), to run the power stage of a power converter, to run the control loop of a digital low dropout regulator (LDO), to monitor a signal with a clocked comparator or the like, among many other possible uses.

In low power (LP) applications including ultra low power (ULP) applications, such as those constrained to less than about 10 microwatts (W), the power overhead due to clock generation can be a significant fraction of the overall power consumption. Thus, generating a clock signal with low power, such as less than about 50 nanowatts (nW), can be of high interest. Actually, low power clock generation may enable many other applications and signal monitoring if the power overhead is low enough. Even in systems in which a clock signal is already present, a clock generator that requires less power than distributing the existent clock may result in an overall power reduction. Generating a clock signal with low power or ultra low power is particularly relevant in many low power applications, such as energy autonomous nodes for Internet of Things (IoT), microcontrollers that remain in stand-by mode for long periods of time, etc.

Conventional solutions typically rely on low bias current generation circuits which still require non-negligible power consumption or rely on reduction of the overall supply voltage, resulting in an output swing that is too small. Conventional solutions suffer either from a small voltage swing (far from nominal supply of the corresponding technology node) or very slow rising/falling edges that cannot directly clock the following gates. Indeed, such slow signals create large shot-through current in subsequent CMOS logic. Consequently, these conventional solutions lead to additional requirements on the clock generator to achieve low power consumption in the rest of the circuit as well, such as level-shifting, edge restoration, etc.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the present invention are illustrated by way of example and are not limited by the accompanying figures. Similar references in the figures may indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

FIG. 1 is a simplified block diagram of a controllable oscillator implemented according to an embodiment of the present disclosure.

FIG. 2 is a schematic diagram of the upper and lower ring oscillators of FIG. 1 implemented according to one embodiment of the present disclosure.

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FIG. 3 is a timing diagram plotting the voltages of nodes of the upper and lower ring oscillators of FIG. 2 according to one embodiment of the present disclosure.

FIG. 4 is a schematic diagram of the amplification circuitry of FIG. 1 implemented according to one embodiment of the present disclosure.

FIG. 5 is a schematic diagram of the buffer circuitry of FIG. 1 implemented according to one embodiment of the present disclosure.

FIG. 6 is a series of timing diagrams plotting aligned voltages of nodes of the amplification circuitry of FIG. 4 and voltage of the OSC signal from the buffer circuitry of FIG. 5 versus time according to one embodiment of the present disclosure.

FIG. 7 is a simplified schematic and block diagram illustrating a controller which may be used as the controller of FIG. 1 for analog control according to one embodiment of the present disclosure.

FIG. 8 is a simplified schematic and block diagram illustrating a controller which may be used as the controller of FIG. 1 for digital control according to another embodiment of the present disclosure.

FIG. 9 is a simplified block diagram of a closed loop configuration using the oscillator of FIG. 1 to improve the accuracy and reduce the spread of the output oscillating signal according to one embodiment of the present disclosure.

FIG. 10 is a schematic diagram of an oscillator implemented according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

A low power controllable oscillator with rail-to-rail output as described herein includes DC-shifted ring oscillators and embedded amplification and buffering which provides a rail-to-rail oscillating signal with fast transitions. The ring oscillators are stacked in series with an intermediate controller that may receive a control signal for controlling frequency. Amplification circuitry is responsive to outputs and other phase-shifted nodes of the ring oscillators to level shift the oscillating signals to a rail-to-rail oscillation signal. Buffer circuitry may be provided to additionally optimize transition times of the rail-to-rail oscillation signal to provide an output oscillating signal with sharp edges. The output voltage signal presents sufficiently fast transition edges so that power consumption of a digital gate connected to its output is minimized by minimizing the gate's shot-through current. The power consumption of an oscillator as described herein is well below the power consumption of any equivalent conventional solution. The oscillator as described herein can be used as a building block within any regular closed-loop oscillator.

FIG. 1 is a simplified block diagram of a controllable oscillator 100 implemented according to an embodiment of the present disclosure. The oscillator 100 includes an upper ring oscillator 102, a lower ring oscillator 104, a controller 106, amplification circuitry 108 including one or more amplification stages, and buffer circuitry 110 including one or more buffer stages. The oscillator 100 is coupled to receive power from an upper supply voltage rail 112 developing an upper supply voltage VDD and a lower supply voltage rail 114 developing a lower supply voltage or reference voltage, such as ground (GND). The upper ring oscillator 102 is coupled between VDD and an upper intermediate node 116 and the lower ring oscillator 104 is coupled between a lower intermediate node 118 and GND.

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The controller **106** is coupled between the upper and lower intermediate nodes **116** and **118** and receives a control signal CTRL. The amplification circuitry **108** and the buffer circuitry **110** are both coupled between VDD and GND. The upper ring oscillator **102** has multiple outputs and the amplifier stage **108** has multiple upper inputs that are coupled together through a set of upper nodes **120** as further described herein. The lower ring oscillator **104** has multiple outputs and the amplifier stage **108** has multiple lower inputs that are coupled together through a set of lower nodes **122** as further described herein.

The upper ring oscillator **102** generates multiple (e.g., three) upper oscillating signals that transition within an upper voltage range (UVR, FIG. 3) between VDD and a voltage of the upper intermediate node **116**, and the lower ring oscillator **104** generates multiple (e.g., three) lower oscillating signals that transition within a lower voltage range (LVR) between a voltage of the lower intermediate node **118** and GND. The current through the upper and lower ring oscillators **102** and **104** is modulated by the controller **106** as controlled by the CTRL signal. The controller **106** may be implemented in any suitable manner, such as one or more transistors or switched capacitors or the like, in which the oscillator **100** may be classified as a voltage-controlled oscillator (VCO), a current-controlled oscillator (CCO), or a digital-controlled oscillator (DCO). By adjusting CTRL (which could be a voltage signal, a current signal, a digital signal, etc.), and thus the supply current through the upper and lower ring oscillators **102** and **104**, the controller **106** determines the corresponding delays which in turn adjusts the oscillation frequency of the upper and lower ring oscillators **102** and **104**. The series connection of the upper and lower ring oscillators **102** and **104** and the controller **106** forces the same current through each of them leading to the same nominal oscillation frequency in steady state. The amplification circuitry **108** converts the upper and lower oscillation signals of the ring oscillators **102** and **104** into a primary oscillation signal POSC that transitions between VDD and GND. The buffer circuitry **110** may be provided to increase or otherwise improve the transitions of the POSC signal to provide the output oscillating signal OSC, which also transitions between VDD and GND.

FIG. 2 is a schematic diagram of the upper and lower ring oscillators **102** and **104** implemented according to one embodiment of the present disclosure. The controller **106**, which is coupled between the intermediate nodes **116** and **118**, is represented as a current source controlled by CTRL. The upper ring oscillator **102** includes P-channel devices **P1**, **P2**, **P3**, **P4**, and **P5** and N-channel devices **N1**, **N2**, **N3**, **N4**, and **N5**. The P-channel devices **P1-P5** each have one current terminal coupled to the voltage rail (VDD) and the N-channel devices **N1-N5** each have one current terminal coupled to the upper intermediate node **116**. The other current terminals of **P1** and **N1** are coupled together at a node **204**, the other current terminals of **P2** and **N2** are coupled together at a node **206**, the other current terminals of **P3** and **N3** are coupled together at a node **208**, the other current terminals of **P4** and **N4** are coupled together at a node **210**, and the other current terminals of **P5** and **N5** are coupled together at a node **202**. The P-channel devices **P1-P5** each have a control terminal coupled to respective control terminals of the N-channel devices **N1-N5** at nodes **202**, **204**, **206**, **208**, and **210**, respectively. In this manner, **P1** and **N1** form a first inverter **203** having an input coupled to node **202** and an output coupled to node **204**, **P2** and **N2** form a second inverter **205** having an input coupled to node **204** and an output coupled to node **206**, **P3** and **N3** form a third inverter

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207 having an input coupled to node **206** and an output coupled to node **208**, **P4** and **N4** form a fourth inverter **209** having an input coupled to node **208** and an output coupled to node **210**, and **P5** and **N5** form a fifth inverter **211** having an input coupled to node **210** and an output coupled to node **202**. The inverters of the upper ring oscillator **102** are coupled in series and also in a ring configuration in which the output of the last inverter **211** is coupled to the input of the first inverter **203** at node **202**, in which node **202** is also an upper output node that provides an upper oscillating signal to an upper input of the amplification circuitry **108**.

The lower ring oscillator **104** is configured in substantially similar manner as the upper ring oscillator **102** and includes P-channel devices **P6**, **P7**, **P8**, **P9**, and **P10** and N-channel devices **N6**, **N7**, **N8**, **N9**, and **N10**. The P-channel devices **P6-P10** each have one current terminal coupled to the lower intermediate node **118** and the N-channel devices **N6-N10** each have one current terminal coupled to the lower supply voltage rail **114** (GND). The other current terminals of **P6** and **N6** are coupled together at a node **214**, the other current terminals of **P7** and **N7** are coupled together at a node **216**, the other current terminals of **P8** and **N8** are coupled together at a node **218**, the other current terminals of **P9** and **N9** are coupled together at a node **220**, and the other current terminals of **P10** and **N10** are coupled together at a node **212**. The P-channel devices **P6-P10** each have a control terminal coupled to respective control terminals of the N-channel devices **N6-N10** at the nodes **212**, **214**, **216**, **218**, and **220**, respectively. In this manner, **P6** and **N6** form a first inverter **213** having an input coupled to node **212** and an output coupled to node **214**, **P7** and **N7** form a second inverter **215** having an input coupled to node **214** and an output coupled to node **216**, **P8** and **N8** form a third inverter **217** having an input coupled to node **216** and an output coupled to node **218**, **P9** and **N9** form a fourth inverter **219** having an input coupled to node **218** and an output coupled to node **220**, and **P10** and **N10** form a fifth inverter **221** having an input coupled to node **220** and an output coupled to node **212**. The inverters of the lower ring oscillator **104** are also coupled in series in a ring configuration in which the output of the last inverter **221** is coupled to the input of the first inverter **213** at node **212**, in which node **212** is also a lower output node that provides a lower oscillating signal to a lower input of the amplification circuitry **108**.

The P-channel devices **P1-P10** and the N-channel devices **N1-N10** are depicted as MOS transistors, each having drain and source terminals as current terminals and gate terminals as control terminals. In this manner, the source terminals of **P1-P5** are coupled to VDD, the source terminals of **P6-P10** are coupled to the lower intermediate terminal **118**, the source terminals of **N1-N5** are coupled to the upper intermediate node **116**, and the source terminals of **N6-N10** are coupled to GND. Each of the MOS transistors **P1-P10** and **N1-N10** are shown with a body diode or bulk connection coupled to its source terminal. It is understood that the bulks of the devices might also be coupled to other nodes such as VDD (in case of the PMOS devices) or to GND (in case of the NMOS devices), in order to achieve a different trade-off between area and performance. The drain terminals of **P1** & **N1**, **P2** & **N2**, **P3** & **N3**, **P4** & **N4**, **P5** & **N5**, **P6** & **N6**, **P7** & **N7**, **P8** & **N8**, **P9** & **N9**, and **P10** & **N10** are coupled together at nodes **204**, **206**, **208**, **210**, **202**, **214**, **216**, **218**, **220**, and **212**, respectively. It is understood that alternative types of transistor devices may be used. A decoupling capacitor **CA** is coupled between VDD and the upper intermediate node **116**, a decoupling capacitor **CB** is coupled between the lower intermediate node **118** and GND, and a

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coupling capacitor CEE is coupled between nodes **202** and **212**. It is noted that the decoupling capacitors CA and CB may be omitted, although such decoupling tends to facilitate synchronization between the upper and lower ring oscillators **102** and **104**.

The capacitor CEE is a floating coupling capacitor that synchronizes the phase and frequency of the ring oscillators **102** and **104**. The capacitance of CEE may be determined by an expected amount of mismatch between the two ring oscillators **102** and **104**. Although in this particular embodiment a single capacitor (e.g., CEE) is used as a coupling mechanism between the upper and the lower oscillators, other coupling devices or circuits (or even more than one) could be used as a synchronization mechanism between the **2** ring oscillators. For example, coupling devices or circuits may be coupled between nodes **204** and **214**, between nodes **206** and **216**, etc. The series connection of the two ring-oscillators **102** and **104** with the controller **106** forces the same current through each leading to the same oscillation frequency in steady state. By adjusting the supply currents of the inverter rings via CTRL, the controller **106** also determines the delay of each of the inverters and, as a result, the oscillation frequency of both of the ring oscillators **102** and **104**. The decoupling capacitors CA and CB are used to achieve voltage stability of the upper and lower intermediate nodes **116** and **118**, which also contributes to synchronize the operation of the ring oscillators **102** and **104**. It is noted that although the ring oscillators **102** and **104** each include a series of 5 cascaded inverters, in alternative embodiments, the number of inverters used to implement the ring oscillators may vary from one embodiment to another as well as the sizing of the devices, as this can be used, for example, to determine the operating frequency of the oscillator.

FIG. 3 is a timing diagram plotting the voltages of nodes **202** (dashed line), **206** (dotted line) and **208** (solid line) of the upper ring oscillator **102** and the voltages of nodes **212** (dashed line), **216** (dotted line) and **218** (solid line) of the lower ring oscillator **104** according to one embodiment of the present disclosure. The voltages of nodes **202**, **206** and **208** each oscillate within the upper voltage range UVR between VDD/2 and VDD, while the voltages of nodes **212**, **216** and **218** each oscillate within the lower voltage range LVR between GND and VDD/2. Thanks to capacitive coupling, the corresponding pair of nodes **202** and **212** oscillate at the same frequency and phase with respect to each other. In the rest of the ring, the corresponding pair of nodes **206** and **216** oscillate at the same frequency and about the same phase with respect to each other (which depends upon any mismatch) but at a delayed phase relative to the pair of nodes **202** and **212**, the corresponding pair of nodes **208** and **218** oscillate at the same frequency and about the same phase with respect to each other but at a delayed phase relative to the pair of nodes **206** and **216**, and the corresponding pair of nodes **210** and **220** oscillate at the same frequency and about the same phase with respect to each other but at a delayed phase relative to the pair of nodes **208** and **218**. This is true in spite of mismatch in gate delays because of the coupling mechanism. Each of the ring oscillators **102** and **104** hence generate several low-voltage clock signals with the same frequencies but different phases at the different inverting stages. Additionally, by stacking them together in the manner shown, the upper and lower ring oscillators **102** and **104** produce copies of the same low-voltage (or even subthreshold) signals at the corresponding nodes, except that UVR is DC-shifted up towards VDD while LVR is DC-shifted down towards GND.

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FIG. 4 is a schematic diagram of the amplification circuitry **108** implemented according to one embodiment of the present disclosure. The amplification circuitry **108** includes P-channel devices **P11**, **P12**, **P13**, **P14**, **P15**, and **P16**, and N-channel devices **N11**, **N12**, **N13**, **N14**, **N15**, and **N16**. The P-channel and N-channel devices are depicted as MOS transistors in a similar manner as the upper and lower ring oscillators **102** and **104**, in which it is understood that alternative types of transistor devices may be used. **P11**, **P13**, and **P15** each have a source terminal and a bulk connection coupled to VDD. **P12** has a source terminal and bulk connection coupled to a drain terminal of **P11** and a drain terminal coupled to a center node **402**. **P14** has a source terminal and bulk connection coupled to a drain terminal of **P13** and a drain terminal coupled to another center node **404**. **P16** has a source terminal and bulk connection coupled to a drain terminal of **P15** and a drain terminal coupled to yet another center node **406** which develops the POSC signal. It is noted that the bulk connections of each of the PMOS transistors may instead be coupled to VDD, and that the bulk connections of each of the NMOS transistors may instead be coupled to GND.

N11, **N13**, and **N15** each have a source terminal and a bulk connection coupled to GND. **N12** has a source terminal and bulk connection coupled to a drain terminal of **N11** and a drain terminal coupled to node **402**. **N14** has a source terminal and bulk connection coupled to a drain terminal of **N13** and a drain terminal coupled to node **404**. **N16** has a source terminal and bulk connection coupled to a drain terminal of **N15** and a drain terminal coupled to node **406**. **P14** and **N14** have their gate terminals coupled together at node **402** and **P16** and **N16** have their gate terminals coupled together at node **404**. **P11** and **P15** have their gate terminals coupled together at node **208** and **N11** and **N15** have their gate terminals coupled together at node **218**. **P12** has its gate terminal coupled to node **202** and **P13** as its gate terminal coupled to node **206**. **N12** has its gate terminal coupled to node **212** and **N13** as its gate terminal coupled to node **216**. It is understood that although in this particular embodiment the same nodes where the capacitor CEE is connected are coupled to the gates of devices **P12** and **N12**, other pairs of nodes (i.e., **204-214**, **206-216**, **208-218**, **210-220**) could be used instead to connect the two ring oscillators with CEE.

The voltage signals developed on nodes **202**, **206**, **208**, **212**, **216**, and **218** are used by the amplification circuitry **108** to produce a "rail-to-rail" output between VDD and GND. Initially, the signals on corresponding nodes **202** and **212** are used to drive **P12** and **N12**, respectfully, of an inverting stage. Because of the initially slow rising and falling edges of the signals on corresponding nodes **202** and **212**, however, a regular inverter, such as only including **P12** and **N12**, may experience large short-circuit or "shoot-through" current. **P11** is inserted between VDD and **P12** and driven by the signal on node **208**, and **N11** is inserted between GND and **N12** and driven by the signal on the corresponding node **218**. These clocked outer devices are added to avoid such large short-circuit current within the first inverting stage. In this manner, time-shifted signals with different phases driving the inverting stage and the clocking devices are used to generate a rail-to-rail signal on node **402** while avoiding any shoot-through current.

The rail-to-rail voltage on node **402** is used to drive a second inverting stage including **P14** and **N14** to develop another rail-to-rail voltage on node **404**. The second inverting stage also includes outer devices **P13** (coupled between VDD and **P14**) and **N13** (coupled between GND and **N14**), in which **P13** is driven by the voltage on node **206** while **N13**

is driven by the voltage on corresponding node **216**. In addition, the rail-to-rail voltage generated on node **404** is used to drive a third inverting stage including **P16** and **N16** to develop POSC as another rail-to-rail voltage on node **406**. The third inverting stage also includes outer devices **P15** (coupled between VDD and **P16**) and **N15** (coupled between GND and **N16**), in which **P15** is driven by the voltage on node **208** while **N15** is driven by the voltage on corresponding node **218**. The second and third inverting stages are each used to further increase the output transition slopes while minimizing the shoot-through current. It is understood that the number of stages in the amplification circuitry **108** can vary from one to as many as it is considered necessary for the generation of the POSC signal, assuming the proper use of the available phase-shifted signals to avoid short-circuit currents. If a finer phase difference is desired between the different signals driving the amplification circuitry, ring oscillators with a larger number of stages, and hence a larger number of available phases, can be used.

FIG. **5** is a schematic diagram of the buffer circuitry **110** implemented according to one embodiment of the present disclosure. Below a certain rise and fall time, long-channel inverters may become more power efficient than their clocked counterpart for a similar gain. For that reason, the buffer circuitry **110** may be provided including a last set of inverting stages which use regular inverters (with longer than minimum channels) to improve the output signal transitions slopes. The buffer circuitry **110** includes P-channel devices **P17**, **P18**, and **P19**, and corresponding N-channel devices **N17**, **N18**, and **N19**. **P17**, **P18**, and **P19** each has its source terminal and bulk connection coupled to VDD, and **N17**, **N18**, and **N19** each has its source terminal and bulk connection coupled to GND. The gate terminals of **P17** and **N17** are coupled together at node **406** for receiving POSC and the drain terminals of **P17** and **N17** are coupled together at a node **408**. The gate terminals of **P18** and **N18** are coupled together at node **408** and the drain terminals of **P18** and **N18** are coupled together at a node **410**. The gate terminals of **P19** and **N19** are coupled together at node **410** and the drain terminals of **P19** and **N19** are coupled together at a node **412** which develops the output oscillation signal OSC. In this manner, **P17** and **N17** are coupled in series between VDD and GND to form a first inverting stage from POSC to node **408**, **P18** and **N18** are coupled in series between VDD and GND to form a second inverting stage from node **408** to node **410**, and **P19** and **N19** are coupled in series between VDD and GND to form a third inverting stage from node **410** to output node **412** providing the oscillating output signal OSC. It is noted that although the buffer circuitry **110** includes 3 stages, it is understood that if buffer circuitry is provided, it may include any number of stages such as 1 or more stages.

FIG. **6** is a series of timing diagrams plotting aligned voltages of nodes **402**, **404**, and **406** (POSC) of the amplification circuitry **108** and voltage of the OSC signal from the buffer circuitry **110** versus time according to one embodiment of the present disclosure. Each of these nodes/voltages oscillate rail-to-rail between VDD and GND in response to being driven by the upper and lower ring oscillators **102** and **104**. With reference back to FIG. **3**, several time points **t0**, **t1** and **t2** are plotted once the upper and lower ring oscillators **102** and **104** stabilize and reach steady state oscillation. These same time points are plotted in each of the timing diagrams of FIG. **5**. The voltage of node **402** at the output of the first inverting stage of the amplification circuitry **108** exhibits smaller slopes and relatively slow transitions. The voltage of node **404** at the output of the second inverting

stage of the amplification circuitry **108** exhibits somewhat larger slopes and faster transitions. The voltage of node **406** at the output of the third inverting stage of the amplification circuitry **108**, which is the POSC signal, exhibits faster and cleaner transitions. The voltage of node **412** at the output of the buffer circuitry **110**, which is the OSC signal, exhibits very fast and very clean transitions. The buffer circuitry **110** may be omitted depending upon the requirements of the particular configuration.

FIG. **7** is a simplified schematic and block diagram illustrating a controller **706** which may be used as the controller **106** for analog control according to one embodiment of the present disclosure. The controller **706** is illustrated as a single MOS transistor device having current terminals coupled between the upper and lower intermediate nodes **116** and **118**. The MOS transistor is shown generically and may be either an NMOS or a PMOS transistor depending upon the particular configuration, with its bulk connection appropriately coupled (e.g., source terminal, VDD GND, etc.). Although only one transistor device is shown, it is understood that it may be implemented with multiple transistor devices coupled in parallel, or any other circuit that modifies the injected current or impedance at the command of an electric signal. Analog control circuitry **702** generates the CTRL as an analog voltage signal provided to the control terminal of the controller **706**, in which CTRL has a voltage level that is varied between a minimum voltage level and a maximum voltage level for adjusting the frequency of the oscillator **100** (hence, a voltage-controlled oscillator or VCO). For example, as the voltage of CTRL is increased, the impedance of the controller **706** decreases thereby increasing the current through the upper and lower ring oscillators **102** and **104** to increase the frequency of operation.

FIG. **8** is a simplified schematic and block diagram illustrating a controller **806** which may be used as the controller **106** for digital control according to another embodiment of the present disclosure. The controller **806** is illustrated as N+1 MOS transistor devices coupled in parallel, each having current terminals coupled between the upper and lower intermediate nodes **116** and **118**. Each of the MOS transistors are shown generically and may be either NMOS or PMOS transistors depending upon the particular configuration, with their bulk connection appropriately coupled (e.g., source terminal, VDD GND, etc.). N may be any suitable integer number greater than zero, and the transistor devices may or may not be equally-sized. Digital control circuitry **802** generates CTRL as digital signal with N+1 individual bits **D1**, **D2**, . . . , **DN**, each provided to the control terminal of a respective one of the N+1 transistor devices of the controller **806**. The controller **806** provides CTRL as a digital value that is varied between a minimum level (e.g., all zero's or vice-versa) and a maximum voltage level (e.g., all one's or vice-versa) for adjusting the frequency of the oscillator **100** (hence, a digital controlled oscillator or DCO). In one embodiment, each bit either turns on or off the corresponding MOS transistor device to adjust the impedance of the controller **806** to adjust the operating frequency. Although each transistor device is shown as a single transistor, it is understood that each may be implemented with multiple transistor devices coupled in parallel, or with any other circuit that modifies the injected current or impedance at the command of an electric signal.

FIG. **9** is a simplified block diagram of a closed loop configuration **900** using the oscillator **100** to improve the accuracy and reduce the spread of the OSC signal according to one embodiment of the present disclosure. Timing refer-

ence circuitry **902** generates and provides a reference time value TREF to one input of a timing comparator **904**. The timing reference circuitry **902** may be implemented as a resistor-capacitor (RC) timing circuit or the like. The timing comparator **904** has an output developing and providing an error signal ERR to an input of a loop filter **906**, which has an output that develops and provides the CTRL signal to the oscillator **100**. The oscillator **100** develops and outputs OSC (or POSC if the buffer circuitry **110** is not included) to an input of a frequency divider **908** and to an input of a loop controller **910**. The frequency divider **908** has an output providing a divided frequency signal to another input of the loop controller **910**, which develops and provides a timing signal TOSC to another input of the timing comparator **904**. The loop controller **910** may also provide a duty cycle control (DCC) signal to the timing reference circuitry **902** for controlling duty cycle. In one embodiment, TREF may represent a target period of OSC while TOSC may represent a corresponding measured period of OSC or POSC.

In operation, the timing comparator **904** compares TREF with TOSC to provide ERR used to adjust CTRL and thus the frequency of OSC (or POSC) until TOSC matches TREF. DCC may be used to adjust duty cycle. The closed loop configuration **900** illustrates an example of using the oscillator **100** within a closed loop to improve accuracy of the frequency of the output OSC (or POSC) signal.

FIG. **10** is a simplified schematic and block diagram of a controllable oscillator **1000** implemented according to another embodiment of the present disclosure. The oscillator **1000** includes an upper ring oscillator **1002**, a lower ring oscillator **1004**, a controller **1006**, and an amplification circuitry **1008**. The upper ring oscillator **1002** is coupled between VDD and an upper intermediate node **1016** and the lower ring oscillator **1004** is coupled between a lower intermediate node **1018** and GND. The controller **1006** is coupled between the upper and lower intermediate nodes **1016** and **1018** and receives a control signal CTRL. The amplification circuitry **1008** is coupled between VDD and GND. The upper ring oscillator **1002** includes inverters **I1**, **I2**, and **I3** coupled in series in a cascade configuration. **I1** has an input coupled to a node **1010** and has an output coupled to an input of **I2**, which has an output coupled to an input of **I3**, which has an output coupled to node **1010** in ring fashion. The lower ring oscillator **1004** includes inverters **I4**, **I5**, and **I6** coupled in series in a cascade configuration. **I4** has an input coupled to a node **1012** and has an output coupled to an input of **I5**, which has an output coupled to an input of **I6**, which has an output coupled to node **1012** in ring fashion. A coupling capacitor CC is coupled between nodes **1010** and **1012**. The amplification circuitry **1008** includes a PMOS transistor PA and an NMOS transistor NA. PA has a source terminal coupled to VDD, a gate terminal coupled to node **1010**, and a drain terminal coupled to an output node **1014** developing POSC. NA has a source terminal coupled to the output node **1014**, a gate terminal coupled to node **1012**, and a drain terminal coupled to GND.

The inverters **I1-I3** each have an upper supply terminal coupled to VDD and a lower supply terminal coupled to the upper intermediate node **1006**. The inverters **I4-I6** each have an upper supply terminal coupled to the lower intermediate node **1018** and a lower supply terminal coupled to GND. In this manner, the inverters **I1-I3** are DC-shifted towards VDD while the inverters **I4-I6** are DC-shifted towards GND in a similar manner described previously for the upper and lower ring oscillators **102** and **104**. The amplification circuitry **1008** converts the DC-shifted upper and lower oscillation signals of the ring oscillators **1002** and **1004** into the

rail-to-rail primary oscillation signal POSC that transitions between VDD and GND in a similar manner as previously described. The controller **1006** may be configured in any suitable manner for voltage, current, or digital control as previously described.

Various modifications are contemplated. The upper and lower ring oscillators **1002** and **1004** may include any odd number of cascaded inverters. Additional coupling devices or circuitry may be included, such as coupled between corresponding intermediate nodes of the inverters **I1-I3** and **I4-I6**. Decoupling capacitors may be included, such as coupled between VDD and node **1016** and between node **1018** and GND, in which such decoupling capacitors may facilitate synchronization between the upper and lower ring oscillators **1002** and **1004**. Although the amplification circuitry **1008** includes only one stage, additional stages may be included, and each stage may include additional cascaded (e.g., stacked) devices in a similar manner as previously described to avoid any direct current path. Buffer circuitry may be provided including one or more buffer stages.

It is further noted that FIGS. **1**, **2** and **10** illustrate embodiments using ring oscillators as the upper and lower oscillators. Ring oscillators provide one or more intermediate nodes facilitating synchronization between the upper and lower oscillators. It is understood, however, that the upper and lower oscillators may be implemented with other types of oscillators in which the present invention is not limited solely to ring oscillators.

Although the present invention has been described in connection with several embodiments, the invention is not intended to be limited to the specific forms set forth herein. On the contrary, it is intended to cover such alternatives, modifications, and equivalents as can be reasonably included within the scope of the invention as defined by the appended claims. For example, variations of positive circuitry or negative circuitry may be used in various embodiments in which the present invention is not limited to specific circuitry polarities, device types or voltage levels or the like. For example, circuitry states, such as circuitry low and circuitry high may be reversed depending upon whether the pin or signal is implemented in positive or negative circuitry or the like. In some cases, the circuitry state may be programmable in which the circuitry state may be reversed for a given circuitry function.

The terms “a” or “an,” as used herein, are defined as one or more than one. Also, the use of introductory phrases such as “at least one” and “one or more” in the claims should not be construed to imply that the introduction of another claim element by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim element to inventions containing only one such element, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an.” The same holds true for the use of definite articles. Unless stated otherwise, terms such as “first” and “second” are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements.

The invention claimed is:

1. A controllable oscillator, comprising:

an upper oscillator coupled between an upper supply voltage and an upper intermediate node that is configured to provide at least one upper oscillating signal on at least one upper oscillating node;

a lower oscillator coupled between a lower intermediate node and a lower supply voltage that is configured to

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provide at least one lower oscillating signal on at least one lower oscillating node; an oscillation controller coupled between the upper intermediate node and the lower intermediate node; and amplification circuitry coupled between the upper supply voltage and the lower supply voltage, having at least one upper input coupled to the at least one upper oscillating node, having at least one lower input coupled to the at least one lower oscillating node, and having a primary output node for providing a primary oscillation signal that oscillates between the upper supply voltage and the lower supply voltage.

2. The controllable oscillator of claim 1, further comprising a coupling circuit coupled between the at least one upper oscillating node and the at least one lower oscillating node.

3. The controllable oscillator of claim 2, wherein the coupling circuit comprises a coupling capacitor.

4. The controllable oscillator of claim 1, wherein the amplification circuitry comprises:

- an upper transistor having current terminals coupled between the upper supply voltage and the primary output node and having a control terminal coupled to the at least one upper oscillating node; and
- a lower transistor having current terminals coupled between primary output node and the lower supply voltage and having a control terminal coupled to the at least one lower oscillating node.

5. The controllable oscillator of claim 1, wherein the upper oscillator comprises a plurality of phase-shifted upper oscillating nodes, and wherein the lower oscillator comprises a plurality of phase-shifted lower oscillating nodes.

6. The controllable oscillator of claim 5, further comprising at least one coupling circuit, each of the at least one coupling circuitry coupled between one of the plurality of phase-shifted upper oscillating nodes and a corresponding one of the plurality of phase-shifted lower oscillating nodes.

7. The controllable oscillator of claim 5, wherein the amplification circuitry comprises:

- an upper P-channel transistor and a lower P-channel transistor having current terminals coupled in series between the upper supply voltage and the primary output node, wherein the upper P-channel transistor has a control terminal coupled to a first one of the phase-shifted upper oscillating nodes, and wherein the lower P-channel transistor has a control terminal coupled to a second one of the phase-shifted upper oscillating nodes; and
- an upper N-channel transistor and a lower N-channel transistor having current terminals coupled in series between the primary output node and the lower supply voltage, wherein the upper N-channel transistor has a control terminal coupled to a first one of the phase-shifted lower oscillating nodes, and wherein the lower N-channel transistor has a control terminal coupled to a second one of the phase-shifted lower oscillating nodes.

8. The controllable oscillator of claim 5, wherein the amplification circuitry comprises a plurality of amplifier stages, each comprising an upper P-channel transistor and a lower P-channel transistor having current terminals coupled in series between the upper supply voltage and a corresponding one of a plurality of center nodes including the primary output node, and an upper N-channel transistor and a lower N-channel transistor having current terminals coupled in series between a corresponding center node and the lower supply voltage.

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9. The controllable oscillator of claim 8, wherein the plurality of amplifier stages comprises a first amplifier stage, comprising:

- a first upper P-channel transistor and a first lower P-channel transistor having current terminals coupled in series between the upper supply voltage and a first center node, and a first upper N-channel transistor and a first lower N-channel transistor having current terminals coupled in series between the first center node and the lower supply voltage; and

wherein the first upper P-channel transistor has a control terminal coupled to a first one of the phase-shifted upper oscillating nodes, wherein the first lower P-channel transistor has a control terminal coupled to a second one of the phase-shifted upper oscillating nodes, wherein the upper N-channel transistor has a control terminal coupled to a first one of the phase-shifted lower oscillating nodes, and wherein the lower N-channel transistor has a control terminal coupled to a second one of the phase-shifted lower oscillating nodes.

10. The controllable oscillator of claim 9, wherein the plurality of amplifier stages comprises a second amplifier stage, comprising:

- a second upper P-channel transistor and a second lower P-channel transistor having current terminals coupled in series between the upper supply voltage and a second center node, and a second upper N-channel transistor and a second lower N-channel transistor having current terminals coupled in series between the second center node and the lower supply voltage; and

wherein the second upper P-channel transistor has a control terminal coupled to a third one of the phase-shifted upper oscillating nodes, wherein the second lower P-channel transistor has a control terminal coupled to the first center node, wherein the second upper N-channel transistor has a control terminal coupled to the first center node, and wherein the second lower N-channel transistor has a control terminal coupled to a third one of the phase-shifted lower oscillating nodes.

11. The controllable oscillator of claim 10, wherein the plurality of amplifier stages comprises a third amplifier stage, comprising:

- a third upper P-channel transistor and a third lower P-channel transistor having current terminals coupled in series between the upper supply voltage and the primary output node, and a third upper N-channel transistor and a third lower N-channel transistor having current terminals coupled in series between the primary output node and the lower supply voltage; and

wherein the third upper P-channel transistor has a control terminal coupled to the first one of the phase-shifted upper oscillating nodes, wherein the third lower P-channel transistor has a control terminal coupled to the second center node, wherein the third upper N-channel transistor has a control terminal coupled to the second center node, and wherein the third lower N-channel transistor has a control terminal coupled to the first one of the phase-shifted lower oscillating nodes.

12. The controllable oscillator of claim 1, further comprising:

- a first decoupling capacitor coupled between the upper supply voltage and the upper intermediate node; and
- a second decoupling capacitor coupled between the lower intermediate node and the lower supply voltage.

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13. The controllable oscillator of claim 1, wherein the oscillation controller comprises a plurality of transistors each having current terminals coupled between the upper and lower intermediate nodes, and each having a control terminal receiving a corresponding one of a plurality of bits of a digital control value.

14. The controllable oscillator of claim 1, wherein the oscillation controller comprises a transistor having current terminals coupled between the upper and lower intermediate nodes and having a control terminal receiving an analog control voltage.

15. The controllable oscillator of claim 1, further comprising buffer circuitry coupled between the upper supply voltage and the lower supply voltage, having an input for coupled to the primary output node and having an output for providing an output oscillating signal, wherein the buffer circuitry is configured to optimize transition times of the output oscillating signal between the upper supply voltage and the lower supply voltage.

16. The controllable oscillator of claim 15, wherein the buffer circuitry comprises a plurality of inverters coupled in series having an input for receiving the primary oscillation signal and having an output for providing the output oscillating signal.

17. The controllable oscillator of claim 16, wherein the plurality of inverters includes a first inverter having an input

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for receiving the primary oscillation signal and an output providing a first oscillation signal, includes a second inverter having an input for receiving the first oscillation signal and an output providing a second oscillation signal, and including a third inverter having an input for receiving the second oscillation signal and having an output for providing a third oscillation signal.

18. The controllable oscillator of claim 16, wherein each the plurality of inverters comprises a P-channel transistor and an N-channel transistor having current terminals coupled in series between the upper supply voltage and the lower supply voltage.

19. The controllable oscillator of claim 1, wherein: the upper oscillator comprises a first plurality of inverters coupled in series in a first ring configuration forming a plurality of upper oscillating nodes; and wherein the lower oscillator comprises a second plurality of inverters coupled in series in a second ring configuration forming a plurality of lower oscillating nodes.

20. The controllable oscillator of claim 19, wherein each of the first and second plurality of inverters comprises a P-channel transistor coupled in series with an N-channel transistor.

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